



Product Discovery AI Agent For E-Commerce Stores

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Phase A — 2026



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Summary, next steps and questions

The Problem

E-commerce search is broken for natural language



Keyword-Only Search

Cannot interpret intent, synonyms, or complex constraints like 'comfortable' or 'marathon-ready'.

Rowley (2000); Madanchian (2024); Luo et al. (2022)



Rigid Filters

Users must learn the store's exact taxonomy before they can find anything relevant.



No Conversational UI

Queries like 'warm jacket under $\approx$ 200' fail silently — the system has no way to handle intent.



AI Hallucination Risk

Generic chatbots invent products not in the store's live inventory, destroying user trust.

de Freitas & Lotufo (2024); Es et al. (2024); Oche et al. (2025)

"comfortable black running shoes under $\approx$ 400" — every human understands this; most store search engines fail entirely.

Background & Alternatives

What shoppers can use today — and where each approach falls short

	Native Store Search	Generic AI Chatbots
Language Flexibility	Low	High
Inventory Grounding	High	Zero / Hallucinates
Store Awareness	Knows its own catalog only	No live store data at all
Conversational	No	Yes



No existing approach combines language flexibility with inventory grounding— that gap is the starting point of this project.

Research Foundations

Literature review, stakeholder interviews, and questionnaires informing system design

Literature Review

- 25+ academic papers across 5 domains
- E-commerce search & product discovery
- Conversational recommender systems
- RAG & LLM query understanding
- Browser extension engineering

Stakeholder Interviews

- 5 domain experts interviewed
- E-commerce & Shopify operations
- Hardware retail industry
- Fashion retail & online catalog
- Biomedical equipment distribution

Questionnaires

- 2 validated instruments (Business + User)
- 15 Likert items each (1–5 scale)
- 5 open-ended qualitative questions
- Grounded in 17 academic frameworks
- TAM, TXAI, UTAUT2, TRI 2.0, DeLone

25+

Academic Sources

5

Domain Experts

2

Questionnaires

30

Survey Items

17

Validated Scales

Target Users & Scope

Shoppers, Merchants, and the Engineering Team — Shopify First



Shoppers

Find relevant products faster without learning rigid store filter structures.



Merchants

Improve discovery and engagement without a custom search integration project.



Engineering Team

Build a reusable multi-site retrieval system validating academic requirements.

These segments and scope are all grounded in the stakeholder interviews and research we conducted.

Project Roadmap

Academic & engineering process — from literature to working system

Research – Phase A

- Literature Review**
20+ papers · 5 topic domains
- Platform Analysis**
3 Shopify stores manually mapped filter params, and variant structures
- Stakeholder Interviews**
*Multiple domains: Fashion, hardware, e-comm, biomedical
15 FRs + 10 NFRs derived*
- Questionnaires**
*30 items · TAM / TXAI / User survey
pre-use assessment of pain, trust, and adoption intent*
- System Design**
data model, module boundaries, Zod contracts, and full tech stack finalized



Development – Phase B

- Base System + Query Parser**
Next.js shell, API layer, DB connection, Parsed Query model with hybrid rule + LLM parsing
- Filter Mapping + Scraping**
One Shopify site manually mapped. Cheerio scraper + normalizer producing shared product card schema
- Ranking + RAG Response**
Attribute relevance scoring, fallback reranking, grounded chat response — no LLM hallucination
- Affiliate + Site Discovery**
Affiliate redirect endpoint, click tracking logs, auto Shopify schema discovery pipeline
- Multi-Site + Hardening**
Multi-site routing, product cache, retry/timeout policies, structured failure logging
- Chrome Extension Deployment**
overlay chat UI, domain detection, full end-to-end contextual search experience

Project Goal

What CartPilot solves and how it stays grounded

GOAL 01

Interpret human intent into a store's filters

Translate fluid natural language into the structured attributes and store-specific filter parameters an e-commerce site actually understands.

```
"warm jacket under $200" → filter.v.price.lte=200
```

GOAL 02

Keep the AI agent grounded — never hallucinate

Ground every answer in products retrieved live from the store, so the agent can explain results but never invents them.

LLM interprets & explains – retrieval provides the facts



Core Project Requirements

3 main functional and non-functional system requirements

</> Functional (FR)

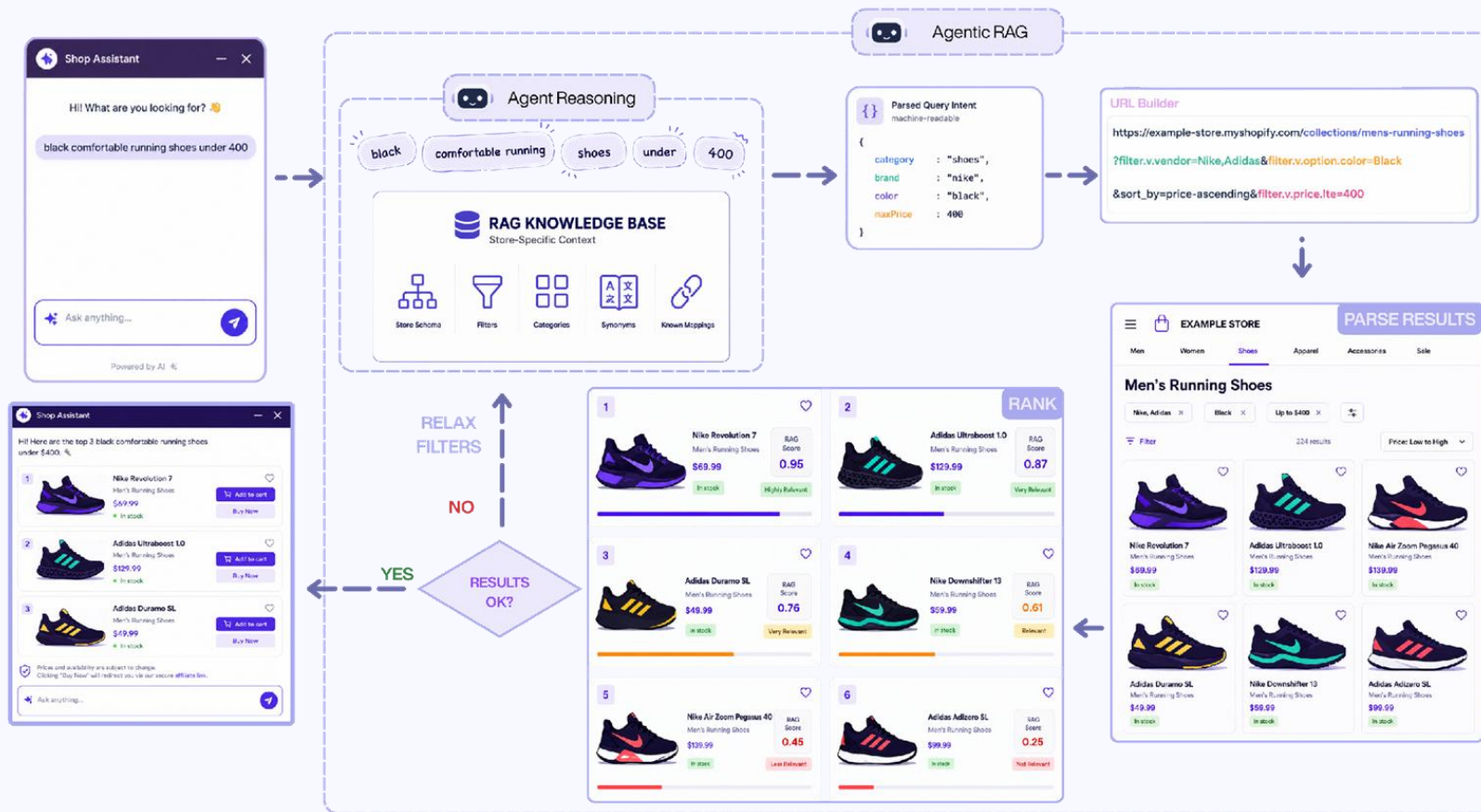
- ✓ **Query Interface & Parsing** Natural Language extraction of category, brand, and constraints via LLM rules.
- ✓ **Store URL Mapping & Scraping** Automated generation of **targeted** storefront URLs and controlled product scraping.
- ✓ **Grounded RAG Responses** Retrieval-Augmented Generation formatting structured, hallucination-free answers.

🔧 Non-Functional (NFR)

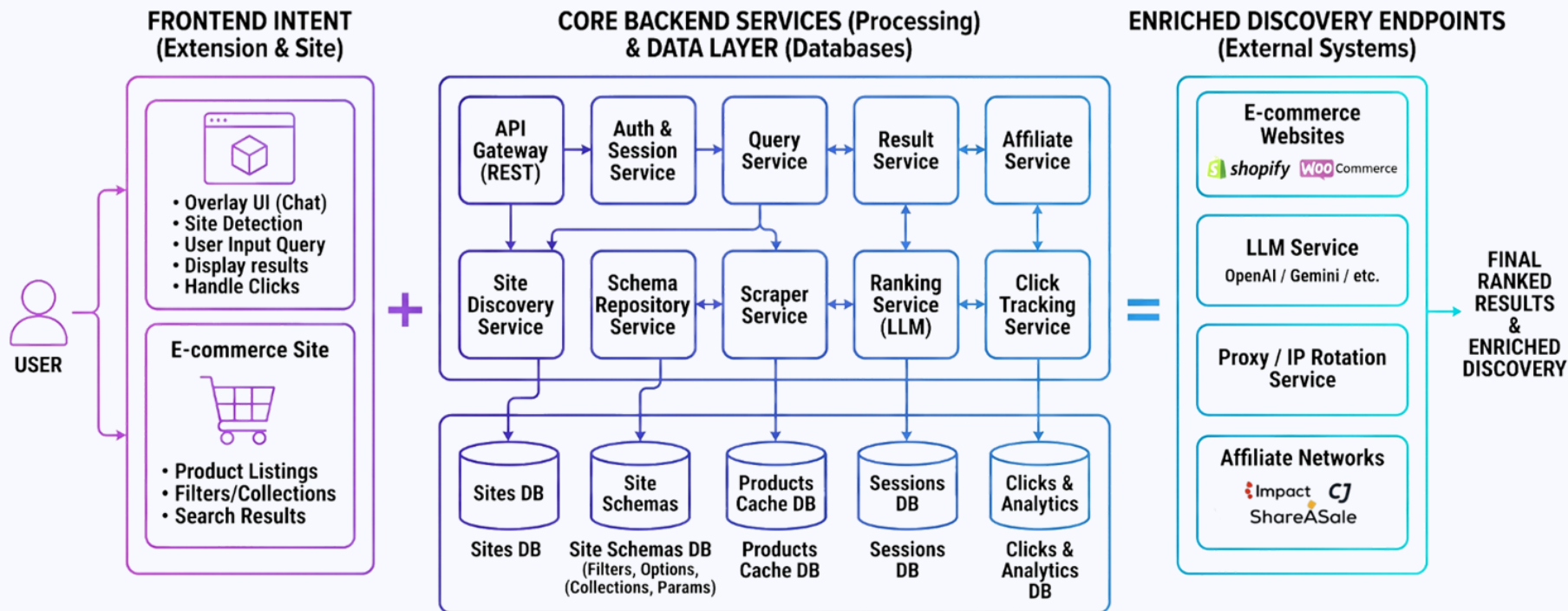
- ✓ **System Grounding (> 90%)** Close to zero hallucinated or unverified product generations.
- ✓ **Latency Targets (< 5s)** Stringent execution times for cached retrieval and query responses.
- ✓ **High Baseline Accuracy** Required parser field accuracy of $\geq 80\%$ per standard benchmark queries.

All of the requirements are derived from academic research and stakeholders interviews

CartPilot — The Solution

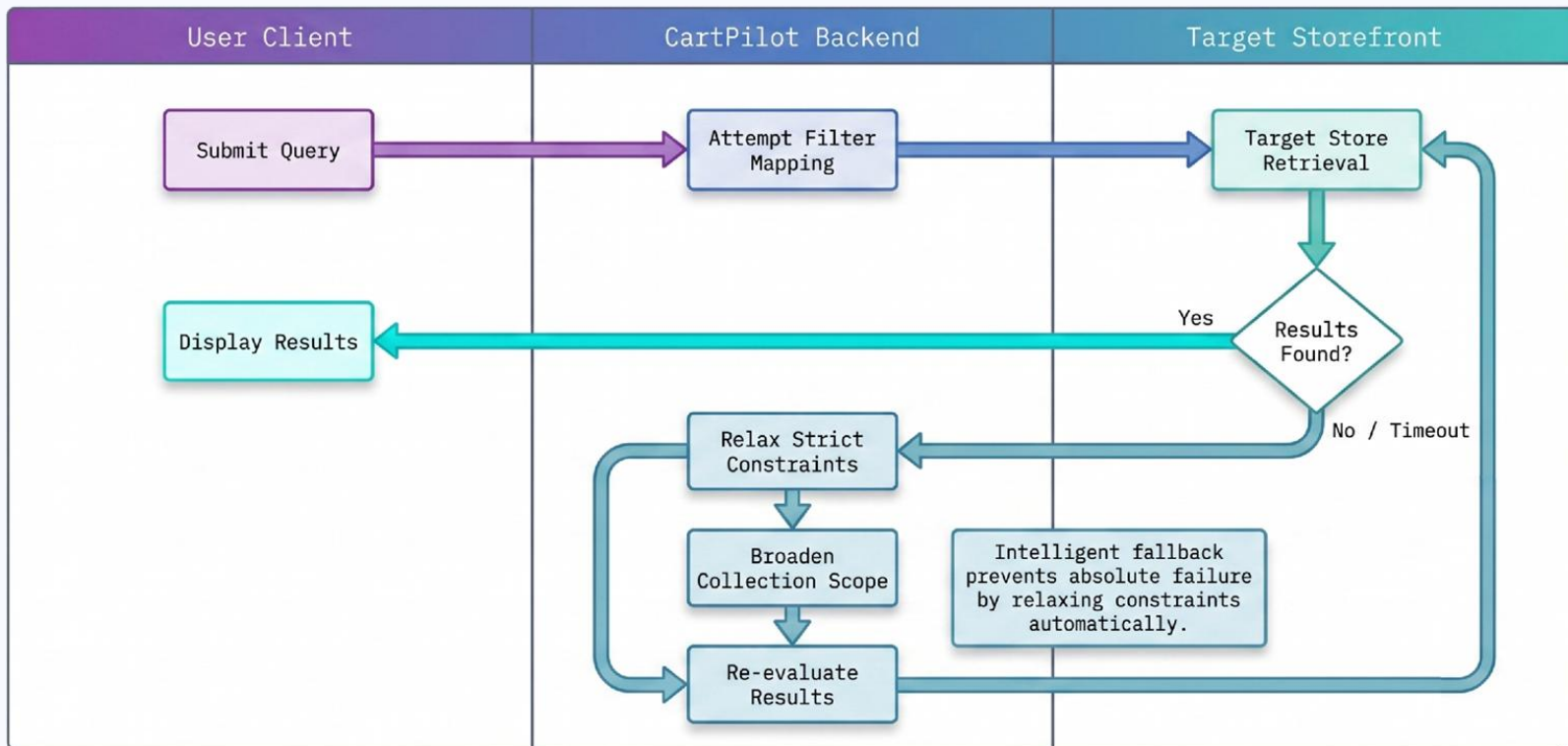


System Architecture & Core Flow

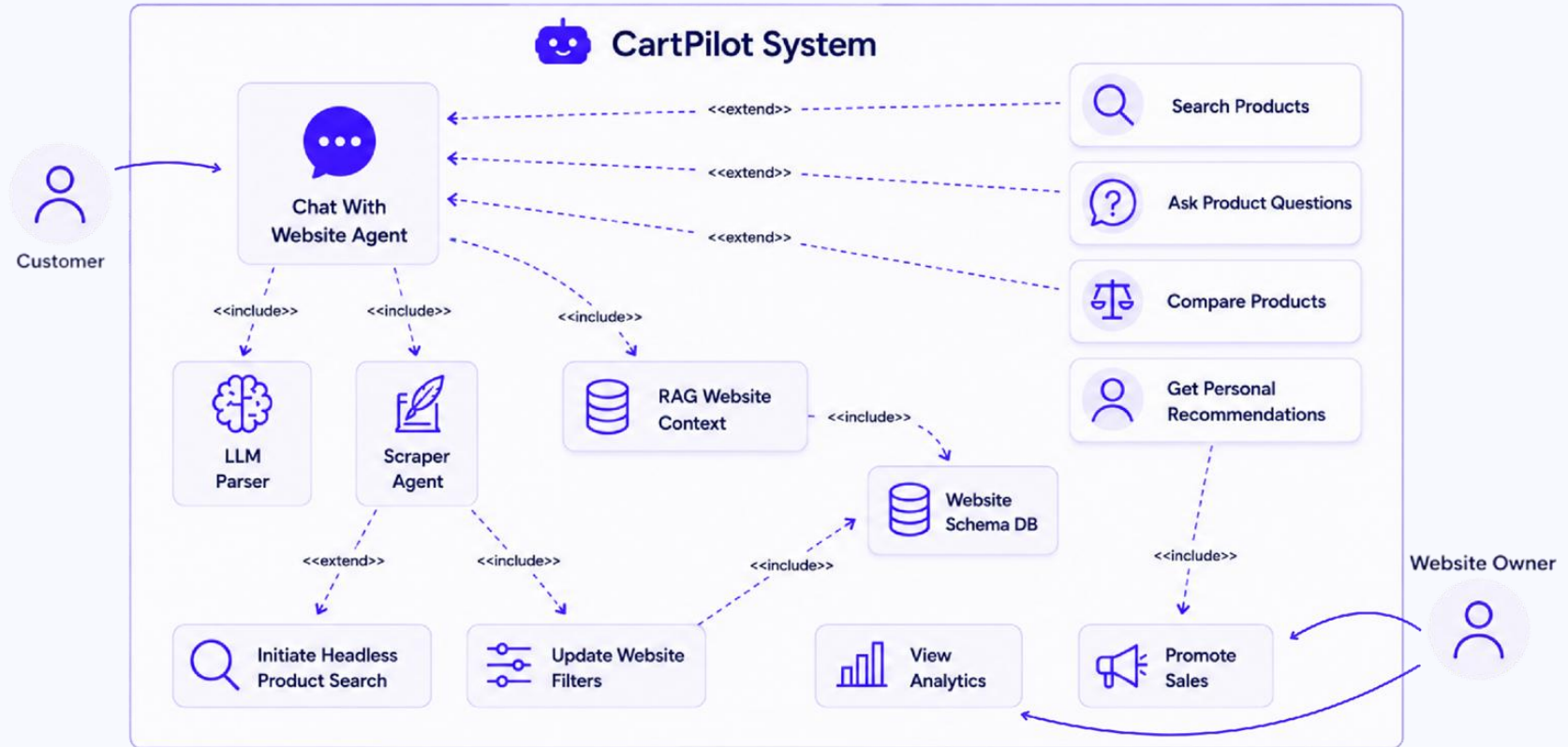


Activity Flow & Fallback Logic

Intelligent fallback prevents absolute failure by relaxing constraints



Use – Case



Success Metrics & Evaluation

9 measurable KPIs defining the success criteria



Accuracy

≥ 80% Parser Field Extraction
Accuracy

≥ 75% Parameter & Filter
Matching

≥ 70% Top-3 Result Relevance



Performance

≤ 5s Cached Query Latency
Threshold

≥ 85% Live Scraping
Success Rate

≥ 70% General Query Success
Rate



Reliability

≥ 95% Safe Fallback on Known
Errors

≥ 95% Affiliate Click-Through
Tracking

30 min Continuous Extension
Stability



Summary & Next Steps

- ✓ Synthesized conversational commerce with structured retrieval
- ✓ Created a robust multi-site schema mapping engine
- ✓ Achieved natural discovery without LLM hallucinations

→ **SaaS Evolution:** Merchant dashboard for analytics and manual schema override

→ **Platform Expansion:** Support for WooCommerce, and other platforms

→ **Semantic Upgrade:** Implementing Vector Indexing for enhanced ranking capabilities

Thank you · Questions?